NETRA: An Encrypted, Offline-Capable LoRa Node for Multi-Hazard Early Warning with Adaptive Baseline Scoring, and a Security and Scaling Analysis

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***Abstract*—Farms and villages without reliable cellular coverage still need early warning of fire, rising water, frost, ground disturbance and poor air. We present NETRA, a low-cost sensing node (ESP32, six sensor modules, SX1278 LoRa at 433.92 MHz) and gateway that scores five hazard indications against an adaptive 10-sample baseline, raises local LCD, buzzer, voice and microSD-log alerts without Internet, and optionally mirrors data to a cloud dashboard over HTTPS. Each 16-byte sensor record is protected with AES-128 before transmission. Rather than claim field accuracy we have not measured, this paper reports what we did measure and compute: (i) an experiment on six packets captured from the running node showing that the gateway decrypts all six correctly, that 768 single-bit ciphertext flips produce on average 63.8 flipped plaintext bits (and only 4 of 768 pass the checksum), that random forgeries pass the 1-byte checksum at 0.378% (theory 0.391%), and that all six captured packets are accepted again when replayed, i.e. the present design has no replay protection; (ii) LoRa time-on-air, duty-cycle and pure-ALOHA delivery analysis showing that the 16-byte binary format cuts air time by 37% relative to our earlier text format, and that a 2 s reporting interval supports only about 5 nodes per channel at $\geq$81% delivery; and (iii) an idealised link-budget estimate of 2.3–8.0 km. A 30-day field evaluation against ground truth is stated as future work with a ready protocol.**

***Index Terms*—LoRa, early warning, AES-128, replay attack, ALOHA, low-cost sensors, rural IoT**

I. Introduction

Floods, forest fires, frost, landslides and poor air quality recur across India, and the last step of early warning, getting a message to the people at risk, is the usual point of failure [1]. Cloud dashboards and SMS assume connectivity that remote farms lack. LoRa gives kilometre-scale links without subscriptions [2]–[4], but a node that must be cheap uses noisy sensors, and fixed single-sensor thresholds then produce false alarms; multi-sensor fusion is the standard remedy [5]–[7].

Two further properties matter for a warning system and are often skipped in student-scale prototypes: a forged or replayed packet must not trigger or suppress an alarm, and the radio must obey the licence-exempt rules. This paper makes four contributions: (1) the NETRA design, including an adaptive baseline scoring rule with explicit equations; (2) an *honest security evaluation* of our AES-128 packet format, including attacks it does *not* stop; (3) an air-time, duty-cycle and ALOHA analysis that bounds how many nodes one gateway channel can serve; (4) a link-budget range estimate, clearly labelled as a model. We do not report field detection accuracy: the field study has not been run, and we state the protocol for it in Section VII.

II. Related Work

Specialist systems detect single hazards well: Dryad's Silvanet detects smouldering wildfire with gas sensors on a LoRaWAN mesh [8], and Amrita's wireless sensor network at Munnar, Kerala, was an early landslide-detection deployment [9], [10]. India's Early Warning Dissemination System uses sirens and mass messaging in coastal districts [11]. These are single-hazard or infrastructure-dependent, and considerably costlier to replicate per site. Offline LoRa alerting is itself established; NETRA claims no novelty there. Its aim is a single cheap node covering five hazard indications with local alerting, encrypted transport, and an analysis of the security and scaling limits that such systems must confront. We have not surveyed the literature exhaustively, so we make no claim of priority.

III. System Design

A. Architecture

Fig. 1 shows the system. The node samples every 2 s: DHT22 (temperature, humidity), MQ-2 (smoke), IR flame sensor, MQ-135 (air quality), JSN-SR04T ultrasonic (water distance) and SW-420 (vibration). A JSN-SR04T 5 V echo is divided to about 3.4 V for the ESP32. The node packs one reading into 16 bytes, encrypts it, and sends it by SX1278

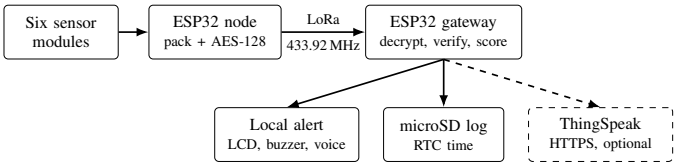


Fig. 1. NETRA architecture. Dashed path is optional.

LoRa [12] at 433.92 MHz. The gateway (ESP32 + SX1278, 16×2 LCD, buzzer, ISD1820 voice, DS3231 RTC, microSD) decrypts, verifies, scores, and alerts. Over Wi-Fi it additionally uploads to ThingSpeak [14] via HTTPS every 15 s and synchronises its RTC by NTP. Sensing, scoring and local alerting need no Internet.

B. Packet Format and Encryption

The record is a packed 16-byte structure (little-endian): zone_id (1 B), checksum (1 B), sequence number (2 B), temperature×100 (2 B, signed), humidity×100 (2 B), smoke ADC (2 B), flame (1 B, 0 = flame), vibration count (1 B), air-quality ADC (2 B), water distance×10 (2 B, signed; -10 marks an invalid reading). The checksum is the XOR of bytes 2–15. The 16 bytes are exactly one AES block and are encrypted with AES-128 in ECB mode (mbedTLS). The gateway rejects any packet that is not 16 bytes, fails the checksum, or has an unknown zone. Section IV analyses what this does and does not protect.

C. Adaptive Baseline and Hazard Scoring

For $x \in \{T, H, S, A, d\}$ the gateway keeps the mean of the last $N = 10$ readings including the current one, $\bar{x}_t = \frac{1}{N}\sum_{i=0}^{N-1} x_{t-i}$. Scoring starts when the window is full (20 s at 2 s). The baseline is frozen while an alert is active, so a sustained event is not absorbed. $N = 10$ was chosen on the bench and not optimised. Let $\mathrm{sat}(u) = \min(\max(u, 0), 1)$, $r^+(x) = (x - \bar{x})/\bar{x}$, $r^-(x) = (\bar{x} - x)/\bar{x}$, $\phi \in \{0, 1\}$ the flame state, n_v the vibration count in the last 60 s, and $\rho_H(k) = \mathrm{sat}(r^+(H)/k)$. The confidences are

$$C_{\text{fire}} = 100[0.50\phi + 0.25\mathrm{sat}(r^+(S)/0.5) + 0.15\mathrm{sat}((T - \bar{T})/5) + 0.10\mathrm{sat}(r^-(H)/0.10)], \quad (1)$$

$$C_{\text{flood}} = 100[0.75 s_w + 0.25\rho_H(0.10)], \quad (2)$$

$$C_{\text{frost}} = 100[0.85 s_T + 0.15\rho_H(0.15)], \quad (3)$$

$$C_{\text{ground}} = 100[0.85\mathrm{sat}(n_v/3) + 0.15\rho_H(0.15)], \quad (4)$$

$$C_{\text{air}} = 100[0.65 s_A + 0.35\mathrm{sat}(r^+(S)/0.3)], \quad (5)$$

with $s_w = 1$ if $0 < d < 25$ cm else $\mathrm{sat}(r^-(d)/0.25)$; $s_T = 1$ for $T \le 2$°C, $(10 - T)/8$ for $2 < T \le 10$, 0 above; $s_A = 1$ if $A > 800$ else $\mathrm{sat}(r^+(A)/0.3)$. The temperature term uses an absolute rise because a relative change of a Celsius value is unstable near zero. An alert is raised when $\max_h C_h \ge \theta = 45$. Weights were assigned by the authors before testing, giving the largest weight to the most specific and directly observable quantity; they are not learned or tuned and this is not machine learning. Because the primary term of each hazard is ≥50 (fire) or higher, at $\theta = 45$ fusion increases sensitivity but cannot suppress a false alarm from a saturated primary sensor (e.g. a flame sensor in sunlight); corroboration is only enforced for $\theta > 100\, w_{\max}$. Water readings that jump more than 40 cm between samples are treated as glitches. Alert starts and clears are written to `ALERTS.txt` on the microSD card with the RTC timestamp, and every reading is appended to a daily CSV.

IV. Security Analysis

Method. We captured six consecutive encrypted packets (seq. 5–10) from the node's serial port together with the AES-128 key used, and executed the gateway's accept test offline in Python (`netra_paper_analysis.py`, provided with this paper).

Correct decryption. All six packets decrypt and pass the checksum, and the decoded values match the node's print-out (T=27.6°C, H=77.8–77.9%, smoke 2910–2928, flame=1 (none), vibration 32–70, AQ 17–34, water 93.9–94.3 cm).

Tampering. Flipping each of the 128 ciphertext bits of each of the six packets (768 trials) changes on average 63.8 of 128 plaintext bits (min 47, max 83), the avalanche behaviour expected of AES, so an attacker cannot make a chosen change to a value (e.g. smoke) without the key. Only 4 of 768 (0.52%) of the garbled results pass the checksum, and those would carry meaningless values.

Forgery. 200,000 random 16-byte blocks were submitted; 756 (0.378%) were accepted, close to the $1/256 = 0.391\%$ expected for a 1-byte checksum. Random junk can therefore occasionally be accepted as a (random) reading. The zone filter and plausibility checks reduce but do not remove this. An 8-byte authentication tag would reduce it to 2^{-64}.

Replay. All 6 of 6 previously captured valid ciphertexts were accepted again. The sequence number is carried but not checked. An attacker who records a normal packet can therefore replay "all clear" during a real hazard, or replay a stale hazard packet; this is the most important weakness of the current design.

Other known weaknesses. (a) ECB encrypts equal blocks to equal ciphertexts; with a sequence number in every block this leaks little, but it is not best practice. (b) Radio jamming cannot be prevented. (c) The key is compiled into firmware in clear; an extracted key compromises every node sharing it. (d) The HTTPS uplink currently does not validate the server certificate (`setInsecure`), so it hides data from passive listeners but not from an active man-in-the-middle. (e) A leaked ThingSpeak write key allows fake data on the dashboard (the local alarm is unaffected).

Remedies (not yet implemented). Accept a packet only if its sequence number exceeds the last accepted one; raise a "node silent" alarm if no valid packet arrives for k intervals; replace AES-ECB plus XOR by an authenticated mode (AES-CCM or GCM, or AES-CMAC) with per-node keys; pin the CA certificate; enable ESP32 flash encryption.

TABLE I
TIME ON AIR (MS), 125 KHZ, CR 4/5

	SF7	SF8	SF9	SF10	SF11	SF12
16 B encrypted	51.5	92.7	164.9	329.7	659.5	1318.9
37 B text	82.2	143.9	267.3	493.6	1069.1	1974.3

TABLE II
PURE-ALOHA DELIVERY (%), SF7, 16-BYTE PACKET

Interval \ nodes	1	5	10	20	50
2 s	100.0	81.4	62.9	37.6	8.0
10 s	100.0	96.0	91.2	82.2	60.4
30 s	100.0	98.6	97.0	93.7	84.5
60 s	100.0	99.3	98.5	96.8	91.9

V. AIR TIME, DUTY CYCLE AND SCALING

A. Time on air

We use the SX1278/AN1200.13 time-on-air formula [12] with SF7–12, 125 kHz, CR 4/5, 8-symbol preamble, explicit header and CRC on (the library defaults, assumed and to be confirmed on the deployed build). Table I compares the 16-byte encrypted binary packet with our earlier 37-byte plain-text format.

At SF7 the binary format reduces air time by 37% (51.5 vs. 82.2 ms). Sent every 2 s, a node occupies the channel 2.57% of the time (4.11% for text), below the 10% duty limit of the Indian 433 MHz licence-exempt rules [13], which also limit transmit power to 10 mW e.r.p. Our bench prototype used the transceiver library's default transmit power (17 dBm), which exceeds that limit; it must be lowered (to $\leq$10 dBm minus antenna gain) before any field use. The 433.05–434.79 MHz rule limits power and duty cycle; we did not find a bandwidth limit in it. The 865–868 MHz band falls under separate 2021 short-range-device rules with different power, bandwidth and duty-cycle limits, which should be checked before choosing it.

B. Channel capacity

With no listen-before-talk, nodes behave as pure ALOHA. The probability that a packet suffers no collision with $n-1$ other nodes is $P = \exp(-2(n-1)\tau/T)$ with τ the air time and T the reporting interval (Table II; Fig. 2a).

This is an upper bound on delivery: it ignores the LoRa capture effect, which helps, and external interference, which hurts. The table shows that the present 2 s interval does not scale: 50 nodes would deliver only 8.0% of packets, whereas a 60 s interval (reasonable for slow hazards) delivers 91.9%. Fire and water-level rise are slow relative to 60 s, so the interval can be raised, with event-triggered fast reporting, rather than adding channels. At SF10 the same 60 s interval with 50 nodes delivers 58.4%, showing that a larger spreading factor buys range at the price of capacity.

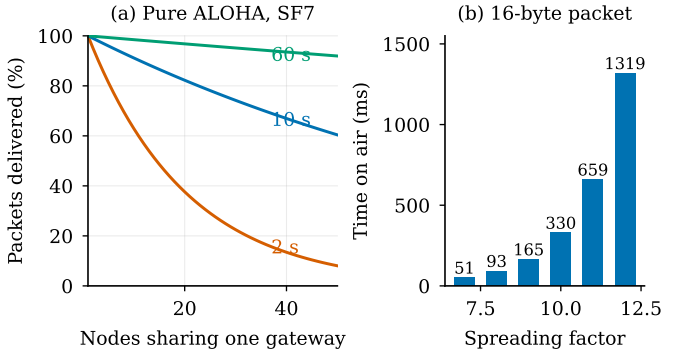


Fig. 2. (a) Pure-ALOHA delivery vs. number of nodes at 2, 10 and 60 s reporting intervals (SF7). (b) Time on air of the 16-byte packet vs. spreading factor.

C. Range estimate (model, not measurement)

Receiver sensitivity is $S = -174 + 10\log_{10}(B) + NF + \mathrm{SNR}_{\mathrm{lim}}$ with $B = 125$ kHz, $NF = 6$ dB, giving -124.5, -132.0 and -137.0 dBm at SF7, SF10 and SF12. With $P_{tx} = 8$ dBm, 0 dBi antennas and a 10 dB fade margin, the maximum path loss is 122.5 dB (SF7) and 130.0 dB (SF10). A two-ray ground-reflection model at 433.92 MHz ($\lambda = 0.691$ m) with a 2 m node gives ranges of 2.31 km (SF7) and 3.56 km (SF10) for a 2 m gateway, and 5.17 km and 7.97 km for a 10 m gateway. These are idealised line-of-sight upper bounds; crops, buildings and terrain will reduce them. They support the design choice of an elevated gateway but are not evidence of achieved range.

VI. BENCH OBSERVATIONS

During bench runs the DHT22 reported 27.6°C when a room thermometer indicated about 21°C, a bias of roughly 6–7°C whose cause (self-heating, placement or sensor offset) we have not diagnosed; we have no calibrated reference and do not correct it. Because the fire term uses a temperature *change* from the rolling baseline, a constant offset does not affect it, but the frost rule uses an absolute value and would. The MQ-135 returned zero until wired to the correct analogue pin and was then 17–80 counts, and MQ sensors need warm-up. An early flood alarm (75% at 156.7 cm) most likely arose from an unstable ultrasonic target on the bench combined with the absence of a debounce; it motivated the 40 cm glitch guard. This guard has a known flaw (it can reject a genuine new level indefinitely) and will be replaced by a persistence rule that accepts a level once three consecutive readings agree within 10 cm.

VII. LIMITATIONS AND FUTURE WORK

No field data exist yet, so we make no claim about precision, recall, false-alarm rate, latency, measured range or battery life. The scoring weights are expert-assigned and untuned. Security gaps are listed in Section IV. The planned evaluation is a 30-day deployment with a ground-truth diary, at least 20 controlled trials per hazard, comparison of fusion against single-sensor thresholds on identical logged data with a

threshold sweep, a packet-delivery/RSSI/SNR range test, and current logging, preceded by implementing replay rejection and authenticated encryption.

VIII. Conclusion

NETRA shows that a single cheap node can score five hazard indications locally, send them encrypted over LoRa, and keep a local event log on microSD. Our analysis shows the AES layer gives confidentiality and defeats controlled tampering, but replay is accepted, the 1-byte checksum admits 0.378% forgeries, and a 2 s reporting interval cannot scale beyond a handful of nodes per channel. These findings, rather than unmeasured accuracy claims, define the next engineering steps.

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